CURRICULUM VITAE

Muhammad Abu Bakar

Geospatial Data Specialist



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Professional experience

11/2025 – present

Geospatial Data Specialist · Beagle Systems, Hamburg

Beagle Systems flies long-range drones beyond visual line of sight (BVLOS) to monitor pipelines, utilities and other infrastructure.

- Developed an open-source drone LiDAR pipeline that georeferences point clouds with PPK, classifies power-line scenes, rebuilds missing conductors from their sag and rates nearby vegetation by risk; tree-fall analysis in progress.
- Built an AI-driven right-of-way and construction-site monitoring pipeline: deep-feature-matching georeferencing, a fine-tuned foundation model for detection, an open-source VLM to verify it, and real-time inference on an NVIDIA Jetson.
- Fine-tuned a building-segmentation foundation model on in-house labels created from 10 cm orthophotos and scaled it across the Cologne area.
- Built a PostGIS and TiTiler backend that serves large GeoTIFFs to the frontend as map tiles, and supported flight planning with pre-flight checks and new geodatabase data.

03/2024 – 10/2025

Geospatial (EO) Data Scientist, part-time · Karuna Technology UG, Germany

- Fine-tuned a pretrained deep-learning model on in-house training data to map urban tree canopies from aerial imagery, then tracked their health with Planet NDVI and NDWI time series in Google Earth Engine to flag underperforming trees.
- Projected the long-term ecosystem services of the mapped trees through a tree-benefits API, from carbon storage to further environmental benefits, estimating the urban canopy's value over the coming decades.
- Built Google Earth Engine workflows that clean multi-year Sentinel-1 radar time series with speckle filtering and apply an equation to derive soil-moisture maps, showing where golf courses suffer from water stress.

04/2021 – 02/2024

GIS Specialist & Data Engineer · MapTransfer, Germany

- Managed the end-to-end geodata workflow for a construction company: georeferenced construction plans and orthophotos, converted them into cloud-optimized GeoTIFFs, uploaded them to AWS S3, and set up QGIS with PostgreSQL as the customer's secure front end.
- Designed and authored several GIS courses for students and professionals, covering LiDAR processing, spatial analysis, Spatial SQL, ALKIS and GeoServer.
- Converted AutoCAD data in QGIS into accurate, ALKIS-compliant GIS layers for telecom infrastructure projects.
- Built interactive Web GIS applications in OpenLayers, serving WMS layers through GeoServer from PostgreSQL.

Education

2017 – 2020

MSc Geology · University of Freiburg, Germany

- Thesis: Simulation of multispectral and hyperspectral imaging techniques for mapping of epithermal ore deposits in vegetated zones.

2016 – 2020	**MSc Remote Sensing & Geographic Information Systems** · National University of Sciences and Technology, Pakistan ◦ Thesis: Development of an open-source Python-based slope stability model that incorporates remote sensing and geotechnical data for slope failure assessment.
2012 – 2015	**BSc Geology** · Bahria University, Pakistan ◦ Thesis: Petrophysical analysis of Joya Mair – Well 4.

Academic projects

Landsat · Sentinel-2	Glacier mass balance of Mýrdalsjökull from historical imagery and DEMs: the melting rate doubled after 2000.
MODIS · TRMM	Snow cover and precipitation variability, 2000–2015.
Landsat 7	Snow-cover change in the Kabul Basin and its impact on water resources.
Landsat 8 · ASTER	Land cover, erosion and morphometry of Mauna Kea.

Further training

2022	NASA Applied Remote Sensing Training (ARSET): five trainings on SAR-based disaster assessment, flood monitoring and modeling, crop mapping with polarimetric SAR and optical data, air-quality model output, and pre- and post-fire monitoring. UNAVCO / GAGE short course: InSAR processing and time-series analysis for geophysical applications (ISCE, ARIA-tools, MintPy).

Skills

AI and machine learning	Fine-tuned foundation models, self-hosted open-source LLMs and VLMs, deep feature matching, gradient boosting; PyTorch, scikit-learn; Hugging Face Transformers, TensorFlow, SAM, XGBoost, LightGBM, CatBoost, vLLM
Vector data	GeoPandas, Shapely, PostGIS; ALKIS, CAD to GIS; buffers, overlays and spatial joins; GeoPackage, Shapefile, GeoJSON; Overpass API
Raster and remote sensing	Rasterio, GDAL, Google Earth Engine; SAR, multispectral and hyperspectral processing; xarray, cloud-optimized GeoTIFFs; ENVI, ERDAS Imagine, eCognition, ESA SNAP
LiDAR	PDAL, CloudCompare, WhiteboxTools; PPK, point-cloud classification; laspy, Open3D, RTKLIB
GIS	QGIS, ArcGIS Pro; publication-ready cartography
Data and cloud	PostgreSQL/PostGIS, DuckDB, AWS (S3, EC2), Docker; SpatiaLite
Web mapping	GeoServer, OpenLayers, TiTiler; Leaflet
Programming	Python, Spatial SQL, R, Bash; Matplotlib, Seaborn, Cartopy; Git
Geology	Numerical modeling, petrophysics

Languages

English	C2
German	B2

Freiburg im Breisgau, 14 October 2026

Signature